

DOACROSS cost-model accuracy — 2 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2088	0.1929	+8.3%	8.3%
Short chain	20,000	1	1	2.5797	2.6071	-1.1%	1.1%
Short, four chains	20,000	4	1	0.9265	1.1851	-21.8%	21.8%
Short, 64 chains	20,000	64	1	0.9442	1.1962	-21.1%	21.1%
Long chain	200,000	1	1	18.2477	23.1653	-21.2%	21.2%
Long, four chains	200,000	4	1	8.0709	11.0317	-26.8%	26.8%
Long, 64 chains	200,000	64	1	7.5373	10.8495	-30.5%	30.5%
Compute in dependency	32,768	1	1	6.0870	7.1094	-14.4%	14.4%
Compute before wait	32,768	4	1	3.4651	3.6269	-4.5%	4.5%
Compute after post	32,768	4	1	3.2820	3.6351	-9.7%	9.7%
Two dependent streams	32,768	4	2	2.4739	3.7596	-34.2%	34.2%
Uneven lanes + compute	8,193	3	1	0.9469	1.0925	-13.3%	13.3%
Large arrays, 64 chains	1,000,000	64	1	41.0913	52.2743	-21.4%	21.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 3 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2636	0.2301	+14.6%	14.6%
Short chain	20,000	1	1	2.4887	2.7993	-11.1%	11.1%
Short, four chains	20,000	4	1	1.0822	1.3689	-20.9%	20.9%
Short, 64 chains	20,000	64	1	0.8898	1.1461	-22.4%	22.4%
Long chain	200,000	1	1	22.8888	26.7877	-14.6%	14.6%
Long, four chains	200,000	4	1	8.3102	10.9524	-24.1%	24.1%
Long, 64 chains	200,000	64	1	10.1434	9.7546	+4.0%	4.0%
Compute in dependency	32,768	1	1	7.1640	7.6575	-6.4%	6.4%
Compute before wait	32,768	4	1	2.7440	2.9948	-8.4%	8.4%
Compute after post	32,768	4	1	2.8986	3.0698	-5.6%	5.6%
Two dependent streams	32,768	4	2	2.5809	3.6256	-28.8%	28.8%
Uneven lanes + compute	8,193	3	1	0.6749	0.6469	+4.3%	4.3%
Large arrays, 64 chains	1,000,000	64	1	33.2851	42.6880	-22.0%	22.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3077	0.2821	+9.1%	9.1%
Short chain	20,000	1	1	2.9890	3.2342	-7.6%	7.6%
Short, four chains	20,000	4	1	0.6719	0.7846	-14.4%	14.4%
Short, 64 chains	20,000	64	1	0.6803	0.8208	-17.1%	17.1%
Long chain	200,000	1	1	26.9862	29.1806	-7.5%	7.5%
Long, four chains	200,000	4	1	5.2551	6.4978	-19.1%	19.1%
Long, 64 chains	200,000	64	1	5.1049	6.5406	-21.9%	21.9%
Compute in dependency	32,768	1	1	7.4508	7.6514	-2.6%	2.6%
Compute before wait	32,768	4	1	1.7648	1.9455	-9.3%	9.3%
Compute after post	32,768	4	1	1.8498	1.9467	-5.0%	5.0%
Two dependent streams	32,768	4	2	1.4599	2.0112	-27.4%	27.4%
Uneven lanes + compute	8,193	3	1	0.6939	0.7437	-6.7%	6.7%
Large arrays, 64 chains	1,000,000	64	1	24.2896	30.2923	-19.8%	19.8%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 5 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4536	0.3609	+25.7%	25.7%
Short chain	20,000	1	1	3.1908	3.3058	-3.5%	3.5%
Short, four chains	20,000	4	1	1.0125	1.0723	-5.6%	5.6%
Short, 64 chains	20,000	64	1	0.8341	0.9869	-15.5%	15.5%
Long chain	200,000	1	1	28.1777	30.9571	-9.0%	9.0%
Long, four chains	200,000	4	1	5.9345	7.8892	-24.8%	24.8%
Long, 64 chains	200,000	64	1	4.5916	5.9092	-22.3%	22.3%
Compute in dependency	32,768	1	1	7.4681	7.8457	-4.8%	4.8%
Compute before wait	32,768	4	1	2.0814	2.3475	-11.3%	11.3%
Compute after post	32,768	4	1	2.0454	2.2958	-10.9%	10.9%
Two dependent streams	32,768	4	2	1.9061	2.6148	-27.1%	27.1%
Uneven lanes + compute	8,193	3	1	0.7432	0.7242	+2.6%	2.6%
Large arrays, 64 chains	1,000,000	64	1	22.6803	30.4167	-25.4%	25.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 6 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.5510	0.4233	+30.2%	30.2%
Short chain	20,000	1	1	3.4456	3.6860	-6.5%	6.5%
Short, four chains	20,000	4	1	0.8785	1.0075	-12.8%	12.8%
Short, 64 chains	20,000	64	1	0.8434	0.8985	-6.1%	6.1%
Long chain	200,000	1	1	30.5549	33.0339	-7.5%	7.5%
Long, four chains	200,000	4	1	4.8519	6.5050	-25.4%	25.4%
Long, 64 chains	200,000	64	1	4.4179	6.0209	-26.6%	26.6%
Compute in dependency	32,768	1	1	7.8163	8.0543	-3.0%	3.0%
Compute before wait	32,768	4	1	1.8577	2.0039	-7.3%	7.3%
Compute after post	32,768	4	1	1.8666	2.0697	-9.8%	9.8%
Two dependent streams	32,768	4	2	1.8025	2.1450	-16.0%	16.0%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	19.0924	26.3218	-27.5%	27.5%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 7 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.5998	0.5060	+18.5%	18.5%
Short chain	20,000	1	1	4.2004	4.0950	+2.6%	2.6%
Short, four chains	20,000	4	1	1.3431	1.3387	+0.3%	0.3%
Short, 64 chains	20,000	64	1	0.8488	0.9371	-9.4%	9.4%
Long chain	200,000	1	1	39.6849	37.7716	+5.1%	5.1%
Long, four chains	200,000	4	1	8.3324	9.2835	-10.2%	10.2%
Long, 64 chains	200,000	64	1	4.1275	5.5654	-25.8%	25.8%
Compute in dependency	32,768	1	1	8.7182	9.3822	-7.1%	7.1%
Compute before wait	32,768	4	1	1.7685	1.8039	-2.0%	2.0%
Compute after post	32,768	4	1	2.0685	2.0908	-1.1%	1.1%
Two dependent streams	32,768	4	2	2.0385	2.4489	-16.8%	16.8%
Uneven lanes + compute	8,193	3	1	1.2430	0.9949	+24.9%	24.9%
Large arrays, 64 chains	1,000,000	64	1	21.4495	25.8933	-17.2%	17.2%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 8 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.7825	0.6607	+18.4%	18.4%
Short chain	20,000	1	1	4.4275	4.6938	-5.7%	5.7%
Short, four chains	20,000	4	1	0.9720	1.1387	-14.6%	14.6%
Short, 64 chains	20,000	64	1	0.6657	0.8394	-20.7%	20.7%
Long chain	200,000	1	1	36.8232	44.4136	-17.1%	17.1%
Long, four chains	200,000	4	1	5.6018	7.5269	-25.6%	25.6%
Long, 64 chains	200,000	64	1	3.0228	4.4453	-32.0%	32.0%
Compute in dependency	32,768	1	1	10.3140	8.9700	+15.0%	15.0%
Compute before wait	32,768	4	1	1.5686	1.7371	-9.7%	9.7%
Compute after post	32,768	4	1	1.6438	1.7470	-5.9%	5.9%
Two dependent streams	32,768	4	2	1.8338	2.2845	-19.7%	19.7%
Uneven lanes + compute	8,193	3	1	1.2996	1.0627	+22.3%	22.3%
Large arrays, 64 chains	1,000,000	64	1	13.3232	18.5635	-28.2%	28.2%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 2 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2087	0.1929	+8.2%	8.2%
Short chain	20,000	1	1	3.7353	2.6071	+43.3%	43.3%
Short, four chains	20,000	4	1	3.6912	1.1851	+211.5%	211.5%
Short, 64 chains	20,000	64	1	3.6826	1.1962	+207.9%	207.9%
Long chain	200,000	1	1	35.4231	23.1653	+52.9%	52.9%
Long, four chains	200,000	4	1	35.3410	11.0317	+220.4%	220.4%
Long, 64 chains	200,000	64	1	35.2460	10.8495	+224.9%	224.9%
Compute in dependency	32,768	1	1	7.3950	7.1094	+4.0%	4.0%
Compute before wait	32,768	4	1	7.8587	3.6269	+116.7%	116.7%
Compute after post	32,768	4	1	7.7558	3.6351	+113.4%	113.4%
Two dependent streams	32,768	4	2	11.4262	3.7596	+203.9%	203.9%
Uneven lanes + compute	8,193	3	1	1.9158	1.0925	+75.4%	75.4%
Large arrays, 64 chains	1,000,000	64	1	176.2559	52.2743	+237.2%	237.2%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 3 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2261	0.2301	-1.7%	1.7%
Short chain	20,000	1	1	2.5217	2.7993	-9.9%	9.9%
Short, four chains	20,000	4	1	2.5474	1.3689	+86.1%	86.1%
Short, 64 chains	20,000	64	1	2.5710	1.1461	+124.3%	124.3%
Long chain	200,000	1	1	23.6258	26.7877	-11.8%	11.8%
Long, four chains	200,000	4	1	23.6524	10.9524	+116.0%	116.0%
Long, 64 chains	200,000	64	1	23.6175	9.7546	+142.1%	142.1%
Compute in dependency	32,768	1	1	5.0961	7.6575	-33.4%	33.4%
Compute before wait	32,768	4	1	5.1915	2.9948	+73.3%	73.3%
Compute after post	32,768	4	1	5.3186	3.0698	+73.3%	73.3%
Two dependent streams	32,768	4	2	7.6866	3.6256	+112.0%	112.0%
Uneven lanes + compute	8,193	3	1	1.4256	0.6469	+120.4%	120.4%
Large arrays, 64 chains	1,000,000	64	1	117.5796	42.6880	+175.4%	175.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2457	0.2821	-12.9%	12.9%
Short chain	20,000	1	1	1.9829	3.2342	-38.7%	38.7%
Short, four chains	20,000	4	1	1.9691	0.7846	+151.0%	151.0%
Short, 64 chains	20,000	64	1	1.9913	0.8208	+142.6%	142.6%
Long chain	200,000	1	1	17.7776	29.1806	-39.1%	39.1%
Long, four chains	200,000	4	1	17.8322	6.4978	+174.4%	174.4%
Long, 64 chains	200,000	64	1	17.7799	6.5406	+171.8%	171.8%
Compute in dependency	32,768	1	1	3.8143	7.6514	-50.1%	50.1%
Compute before wait	32,768	4	1	3.9776	1.9455	+104.4%	104.4%
Compute after post	32,768	4	1	3.9930	1.9467	+105.1%	105.1%
Two dependent streams	32,768	4	2	5.8673	2.0112	+191.7%	191.7%
Uneven lanes + compute	8,193	3	1	1.0996	0.7437	+47.9%	47.9%
Large arrays, 64 chains	1,000,000	64	1	87.9496	30.2923	+190.3%	190.3%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 5 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3297	0.3609	-8.6%	8.6%
Short chain	20,000	1	1	1.6794	3.3058	-49.2%	49.2%
Short, four chains	20,000	4	1	1.6944	1.0723	+58.0%	58.0%
Short, 64 chains	20,000	64	1	1.6996	0.9869	+72.2%	72.2%
Long chain	200,000	1	1	14.3250	30.9571	-53.7%	53.7%
Long, four chains	200,000	4	1	14.2974	7.8892	+81.2%	81.2%
Long, 64 chains	200,000	64	1	14.3408	5.9092	+142.7%	142.7%
Compute in dependency	32,768	1	1	3.1554	7.8457	-59.8%	59.8%
Compute before wait	32,768	4	1	3.3945	2.3475	+44.6%	44.6%
Compute after post	32,768	4	1	3.3174	2.2958	+44.5%	44.5%
Two dependent streams	32,768	4	2	4.8085	2.6148	+83.9%	83.9%
Uneven lanes + compute	8,193	3	1	0.9812	0.7242	+35.5%	35.5%
Large arrays, 64 chains	1,000,000	64	1	70.8149	30.4167	+132.8%	132.8%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 6 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3977	0.4233	-6.0%	6.0%
Short chain	20,000	1	1	1.4956	3.6860	-59.4%	59.4%
Short, four chains	20,000	4	1	1.5040	1.0075	+49.3%	49.3%
Short, 64 chains	20,000	64	1	1.5255	0.8985	+69.8%	69.8%
Long chain	200,000	1	1	12.0227	33.0339	-63.6%	63.6%
Long, four chains	200,000	4	1	12.0062	6.5050	+84.6%	84.6%
Long, 64 chains	200,000	64	1	12.1225	6.0209	+101.3%	101.3%
Compute in dependency	32,768	1	1	2.7201	8.0543	-66.2%	66.2%
Compute before wait	32,768	4	1	2.9622	2.0039	+47.8%	47.8%
Compute after post	32,768	4	1	2.8582	2.0697	+38.1%	38.1%
Two dependent streams	32,768	4	2	4.1112	2.1450	+91.7%	91.7%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	58.8509	26.3218	+123.6%	123.6%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 7 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3833	0.5060	-24.3%	24.3%
Short chain	20,000	1	1	1.3795	4.0950	-66.3%	66.3%
Short, four chains	20,000	4	1	1.3443	1.3387	+0.4%	0.4%
Short, 64 chains	20,000	64	1	1.3744	0.9371	+46.7%	46.7%
Long chain	200,000	1	1	10.4128	37.7716	-72.4%	72.4%
Long, four chains	200,000	4	1	10.4323	9.2835	+12.4%	12.4%
Long, 64 chains	200,000	64	1	10.3734	5.5654	+86.4%	86.4%
Compute in dependency	32,768	1	1	2.4423	9.3822	-74.0%	74.0%
Compute before wait	32,768	4	1	2.4939	1.8039	+38.3%	38.3%
Compute after post	32,768	4	1	2.5635	2.0908	+22.6%	22.6%
Two dependent streams	32,768	4	2	3.6082	2.4489	+47.3%	47.3%
Uneven lanes + compute	8,193	3	1	0.9408	0.9949	-5.4%	5.4%
Large arrays, 64 chains	1,000,000	64	1	50.7437	25.8933	+96.0%	96.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 8 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4517	0.6607	-31.6%	31.6%
Short chain	20,000	1	1	1.3006	4.6938	-72.3%	72.3%
Short, four chains	20,000	4	1	1.2888	1.1387	+13.2%	13.2%
Short, 64 chains	20,000	64	1	1.3149	0.8394	+56.6%	56.6%
Long chain	200,000	1	1	9.2115	44.4136	-79.3%	79.3%
Long, four chains	200,000	4	1	9.2207	7.5269	+22.5%	22.5%
Long, 64 chains	200,000	64	1	9.2202	4.4453	+107.4%	107.4%
Compute in dependency	32,768	1	1	2.2464	8.9700	-75.0%	75.0%
Compute before wait	32,768	4	1	2.3999	1.7371	+38.2%	38.2%
Compute after post	32,768	4	1	2.3122	1.7470	+32.4%	32.4%
Two dependent streams	32,768	4	2	3.2621	2.2845	+42.8%	42.8%
Uneven lanes + compute	8,193	3	1	0.8485	1.0627	-20.2%	20.2%
Large arrays, 64 chains	1,000,000	64	1	44.2396	18.5635	+138.3%	138.3%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane computation and waits; width-specific profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.